4. Selecting an Item Number that Does Not Exist

The customer selects a number to vend that does not exist

Objective	When the customer selects their item, because the item is not real the display should display text requesting a different value or item be selected (Invalid Item)	
Results	The text or alert that the item they chose does not exist and that they need to choose a valid item	
Actors	Basic understanding of how the system works, but other than that it should hopefully be straight forward	
Related use cases	Buying an item, Receiving change after purchasing an item, Not buying an Item, Putting in Values for Items	
Trigger	Punching in an invalid item number	
Stakeholders	The owner of the machine is interested in this as well as the customer who is buying an item from the machine	
Resources	Quartus and the SoC Series V Board	
Assumptions	A working setup with the seven-segment display, working money switches, working RAM set up for items and the working keypad as well	
Post-condition	The system itself shouldn't change, but the 7-segment should display the warning alert for the invalid item	
Frequency	I would be hopeful that this never has to happen, but it really is customer dependent.	
Mission Priority	Again, tied directly to frequency. It depends on the customer and there are no true specification of this for our design, but when you do not tie to a correct value, you should be able to make this work and practically it would be very useful	
Safety Priority	I hope there is no safety concern	
Flow of events	**Step #**	**Action**
	1.	Customer browses the catalog of items
	2.	Customer chooses item that they want
	3.	Value is not a valid item number, and alert appears telling the customer to choose a new item

Hazard/ Result /Mitigation	• If this case is done incorrectly, the entire system is likely incorrect or not working to specifications
Requirements from Use Case	• Display the alert message when the wrong item number is imputted